Time series

EC 421, Set 7

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Prologue

Schedule

Last Time

Asymptotics, probability limits, and consistency

Today

Time series

Soon

- midterm **review** (Tuesday; bring questions)
- midterm

Asymptotics and consistency

Review

Asymptotics and consistency

Review

1. Compare/contrast the concepts *expected value* and *probability limit*.
2. What does it mean if the estimator $\hat{\theta}$ is consistent for θ?
3. What is required for an omitted variable to bias OLS estimates of β_j?
4. Does omitted-variable bias affect the consistency of OLS for β_j?
5. What can we know about the direction of omitted-variable bias?
6. How does measurement error in an explanatory variable affect the OLS estimate for that variable's effect on the outcome variable?
7. How does measurement error in an outcome variable affect OLS?

Time-series data

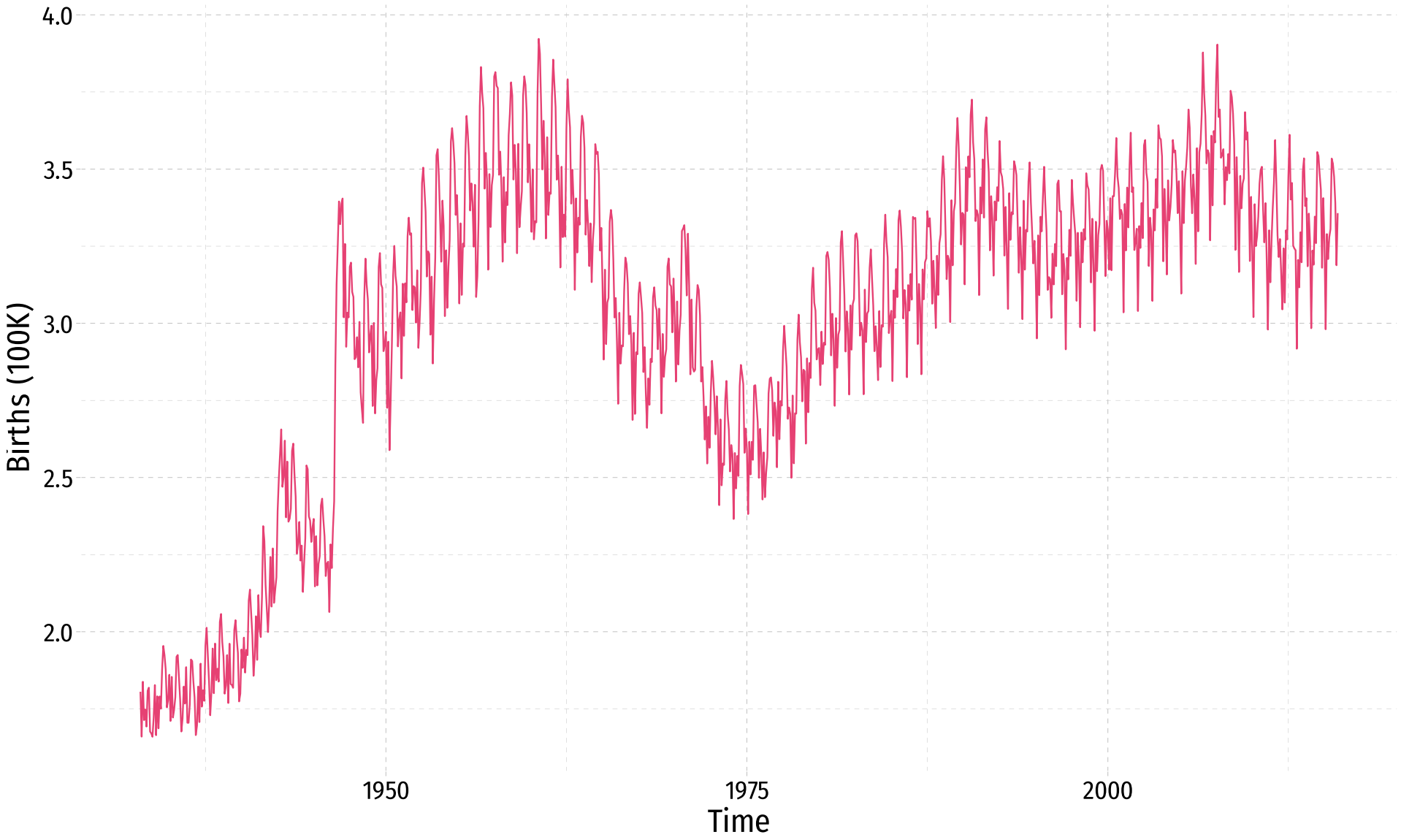
Time-series data

Introduction

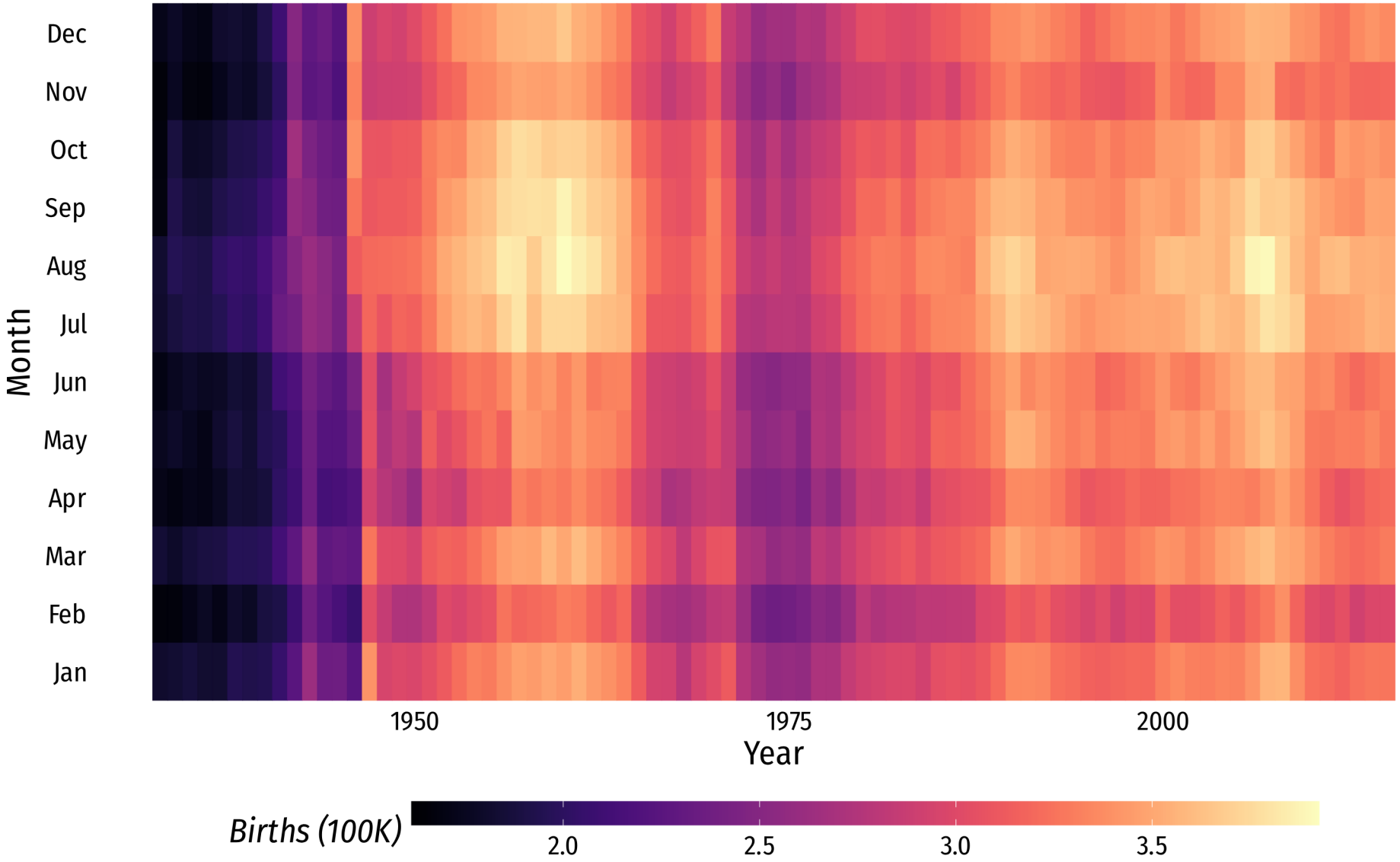
Up to this point, we focused on **cross-sectional data**.

- Sampled *across* a population (*e.g.*, people, counties, countries).
- Sampled at *one moment* in time (*e.g.*, Jan. 1, 2015).
- We had n *individuals*, each indexed i in $\{1, \ldots, n\}$.

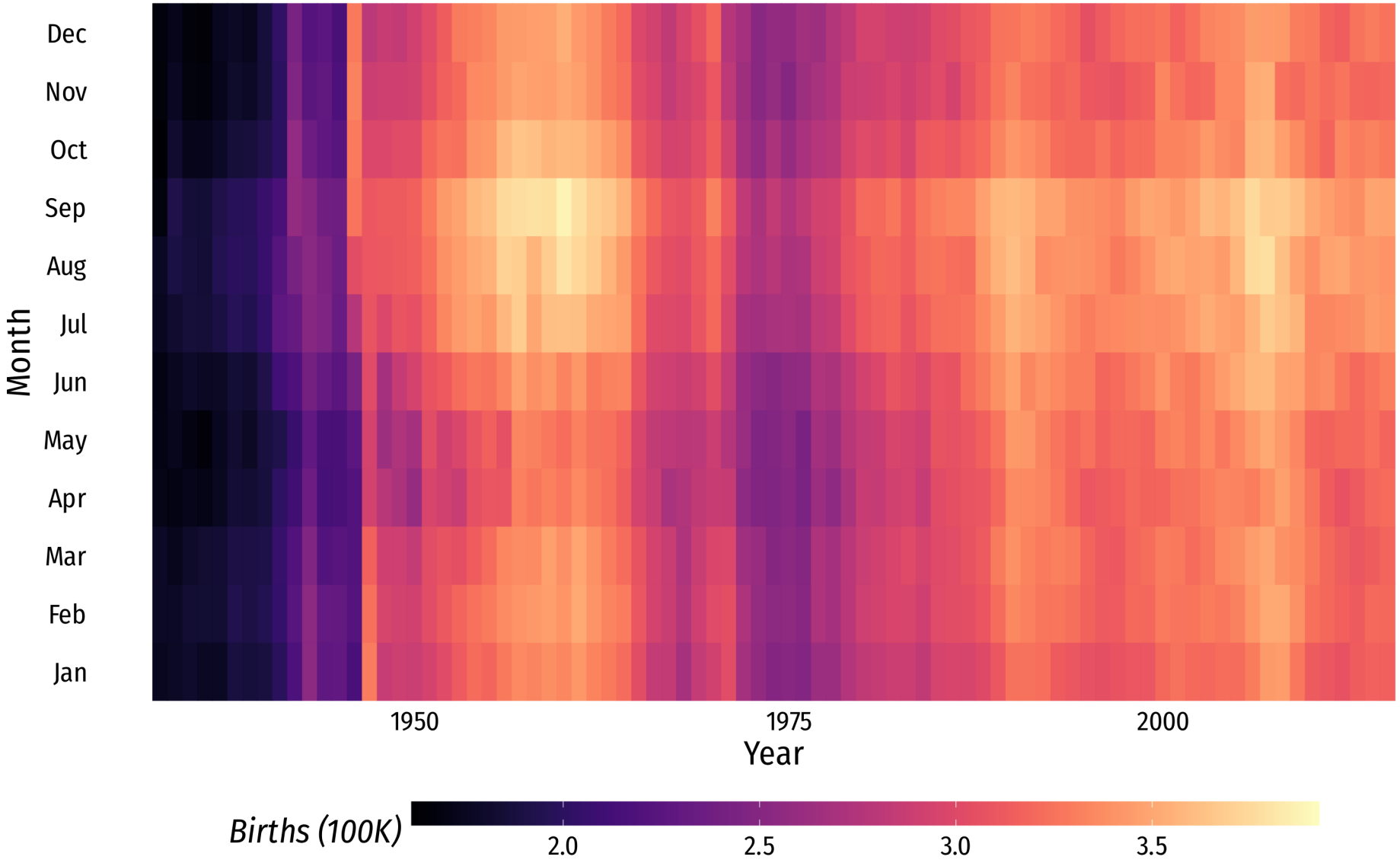
US monthly births, 1933–2015: Classic time-series graph



US monthly births, 1933–2015: Newfangled time-series graph



US monthly births per 30 days, 1933–2015: Newfangled time-series graph



Time-series models

Time-series models

Introduction

Our model now looks something like

$$\text{Births}_t = \beta_0 + \beta_1 \text{Income}_t + u_t$$

Time-series models

Assumptions

1. **New: Weakly persistent outcomes**—essentially, x_{t+k} in the distant period $t+k$ is weakly correlated with period x_t (when k is "big").
2. y_t is a **linear function** of its parameters and disturbance.
3. There is **no perfect collinearity** in our data.
4. The u_t have conditional mean of zero (**exogeneity**), $\boldsymbol{E}\left[u_t|X\right]=0$.
5. The u_t are **homoskedastic** with **zero correlation** between u_t and u_s, *i.e.*, $\text{Var}\left(u_t|X\right)=\text{Var}\left(u_t\right)=\sigma^2$ and $\text{Cor}\left(u_t,\,u_s|X\right)=0$.
6. **Normality of disturbances**, *i.e.*, $u_t \overset{\text{iid}}{\sim} N\left(0,\,\sigma^2\right)$.

(We already know that we can relax the last two assumptions.)

Time-series models

Model options

Time-series modeling boils down to two classes of models.

1. **Static models:** Do not allow for persistent effect.
2. **Dynamic models:** Allow for persistent effects.
    - Models with **lagged explanatory** variables
    - **Autoregressive, distributed-lag** (ADL) models

Time-series models

Model options

Option 1: Static models

Static models assume the outcome depends upon only the current period.

$$\text{Births}_t = \beta_0 + \beta_1 \text{Income}_t + u_t$$

Time-series models

Model options

Option 2: Dynamic models

Dynamic models allow the outcome to depend upon other periods.

Time-series models

Model options

Option 2a: Dynamic models with lagged explanatory variables

The outcome can depend on explanatory variable(s) in other periods.

$$\begin{aligned}\text{Births}_t =& \beta_0 + \beta_1 \text{Income}_t + \beta_2 \text{Income}_{t-1} + \\ & \beta_3 \text{Income}_{t-2} + \beta_4 \text{Income}_{t-3} + u_t\end{aligned}$$

Time-series models

Model options

Option 2b: Autoregressive distributed-lag (ADL) models

The outcome depends on the explanatory variable(s) and/or the outcome variable in prior periods.

$$\text{Births}_t = \beta_0 + \beta_1 \text{Income}_t + \beta_2 \text{Income}_{t-1} + \beta_3 \text{Births}_{t-1} + u_t$$

Autoregressive distributed-lag models

Numbers of lags

ADL models are often specified as $\mathbf{ADL}(p, q)$, where

- p is the (maximum) number of **lags** for the outcome variable.
- q is the (maximum) number of **lags** for explanatory variables.

Autoregressive distributed-lag models

Complexity

Due to their lags, ADL models actually estimate even more complex relationships than you might first guess.

Autoregressive distributed-lag models

Complexity

Continuing...

$$
\begin{aligned}
\text{Births}_t =& \beta_0 + \beta_1 \text{Income}_t + \\
& \beta_2 \underbrace{(\beta_0 + \beta_1 \text{Income}_{t-1} + \beta_2 \text{Births}_{t-2} + u_{t-1})}_{\text{Births}_{t-1}} + u_t \\
=& \beta_0 (1 + \beta_2) + \beta_1 \text{Income}_t + \beta_1 \beta_2 \text{Income}_{t-1} + \\
& \beta_2^2 \text{Births}_{t-2} + u_t + \beta_2 u_{t-1}
\end{aligned}
$$

Autoregressive distributed-lag models

Complexity

Eventually we arrive at

$$
\begin{aligned}
\text{Births}_t = &\beta_0 \left(1 + \beta_2 + \beta_2^2 + \beta_2^3 + \cdots\right) + \\
&\beta_1 \left(\text{Income}_t + \beta_2 \text{Income}_{t-1} + \beta_2^2 \text{Income}_{t-2} + \cdots\right) + \\
&u_t + \beta_2 u_{t-1} + \beta_2^2 u_{t-2} + \cdots
\end{aligned}
$$

1. These lags enter into the equation in a very specific way—not the most flexible specification.

Autoregressive distributed-lag models

Example: The partial-adjustment model

There are times that actually want to model an individual's **desired amount**, rather than her ***actual* amount**

Autoregressive distributed-lag models

Example: The partial-adjustment model

We want to know how the **desired number of cigarettes**, $\widetilde{\mathrm{Cig}}_t$, changes with the current period's cigarette tax, *e.g.*,

$$\widetilde{\mathrm{Cig}}_t = \beta_0 + \beta_1 \mathrm{Tax}_t + u_t \tag{A}$$

Autoregressive distributed-lag models

Example: The partial-adjustment model

$$\widetilde{\text{Cig}}_t = \beta_0 + \beta_1 \text{Tax}_t + u_t \tag{A}$$

$$\text{Cig}_t = \lambda \widetilde{\text{Cig}}_t + (1 - \lambda) \text{Cig}_{t-1} \tag{B}$$

OLS in time series

OLS in time series

Unbiased coefficients

As before, the unbiased-ness of OLS is going to depend upon our exogeneity assumption, *i.e.*, $\boldsymbol{E}\left[u_t|X\right] = 0$.

OLS in time series

Unbiased coefficients

We need both parts of our exogeneity assumption for OLS to be unbiased:

$$\boldsymbol{E}\left[\hat{\beta}_1 \middle| X\right] = \beta_1 + \boldsymbol{E}\left[\frac{\sum_t (x_t - \overline{x}) u_t}{\sum_t (x_t - \overline{x})^2} \middle| X\right]$$

I.e., to guarantee the numerator equals zero, we need $\boldsymbol{E}\left[u_t | X\right] = 0$, for both $\boldsymbol{E}\left[u_t | X_t\right] = 0$ *and* $\boldsymbol{E}\left[u_t | X_s\right] = 0$ $(s \neq t)$.

OLS in time series

Unbiased coefficients

To see why dynamic models with lagged outcome variables violate our exogeneity assumption, consider two periods of our simple ADL(1, 0) model.

$$\text{Births}_t = \beta_0 + \beta_1 \text{Income}_t + \beta_2 \text{Births}_{t-1} + u_t \tag{1}$$

$$\text{Births}_{t+1} = \beta_0 + \beta_1 \text{Income}_{t+1} + \beta_2 \text{Births}_t + u_{t+1} \tag{2}$$

OLS in time series

Consistent coefficients

All is not lost.

OLS in time series

Consistent coefficients

To see why OLS is consistent with contemporaneous exogeneity, consider the OLS estimate for β_1 in

$$\text{Births}_t = \beta_0 + \beta_1 \text{Births}_{t-1} + u_t$$

which we've shown (a few times) can be written

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_t \left(\text{Births}_{t-1} - \overline{\text{Births}}\right) u_t}{\sum_t \left(\text{Births}_{t-1} - \overline{\text{Births}}\right)^2}$$

OLS in time series

Consistent coefficients

$$
\begin{aligned}
\operatorname{plim} \hat{\beta}_1 &= \operatorname{plim} \left(\beta_1 + \frac{\sum_t \left(\text{Births}_{t-1} - \overline{\text{Births}} \right) u_t}{\sum_t \left(\text{Births}_{t-1} - \overline{\text{Births}} \right)^2} \right) \\
&= \beta_1 + \frac{\operatorname{plim} \left[\sum_t \left(\text{Births}_{t-1} - \overline{\text{Births}} \right) u_t / T \right]}{\operatorname{plim} \left[\sum_t \left(\text{Births}_{t-1} - \overline{\text{Births}} \right)^2 / T \right]} \\
&= \beta_1 + \frac{\text{Cov}\left(\text{Births}_{t-1},\, u_t \right)}{\text{Var}\left(\text{Births}_t \right)}
\end{aligned}
$$

OLS in time series

Consistent coefficients

Thus, if we assume **contemporaneous exogeneity**, **OLS is consistent** for the coefficients, *even for models with lagged dependent variables*.

The end.

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Time series

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   - Partial-adjustment models
5. Unbiasedness of OLS
6. Consistency of OLS
7. *Extra:* ADL in equilibrium

Autoregressive distributed-lag models

Equilibrium effects

ADL models also offer interesting insights for long-run/equilibrium effects.

$$\mathbf{Births}_t = \beta_0 + \beta_1 \mathbf{Income}_t + \beta_2 \mathbf{Births}_{t-1} + u_t$$

In this ADL(1, 0) model, β_1 gives the **short-run effect** of income on the number of births.

Autoregressive distributed-lag models

Equilibrium effects

Starting with

$$\text{Births}_t = \beta_0 + \beta_1 \text{Income}_t + \beta_2 \text{Births}_{t-1} + u_t$$

Autoregressive distributed-lag models

Equilibrium effects

Short-run effect of income on births:

$$\text{Births}_t = \beta_0 + \beta_1 \text{Income}_t + \beta_2 \text{Births}_{t-1} + u_t$$

Long-run effect of income on births:

$$\text{Births}^\star = \frac{\beta_0}{(1-\beta_2)} + \frac{\beta_1}{(1-\beta_2)} \text{Income}^\star$$

Autoregressive distributed-lag models

Equilibrium effects

Another way to see this result:

We already showed

$$\text{Births}_t = \beta_0 + \beta_1 \text{Income}_t + \beta_2 \text{Births}_{t-1}$$

gives us

$$\begin{aligned}\text{Births}_t = &\beta_0 \left(1 + \beta_2 + \beta_2^2 + \beta_2^3 + \cdots\right) + \\ &\beta_1 \left(\text{Income}_t + \beta_2 \text{Income}_{t-1} + \beta_2^2 \text{Income}_{t-2} + \cdots\right) + \\ &u_t + \beta_2 u_{t-1} + \beta_2^2 u_{t-2} + \cdots\end{aligned}$$

Autoregressive distributed-lag models

Equilibrium effects

Substituting $\mathbf{Income}_t = \mathbf{Income}^\star$ for all k
(and assuming no disturbances in equilibrium):

Autoregressive distributed-lag models

Equilibrium effects

Substituting $\mathbf{Income}_t = \mathbf{Income}^\star$ for all k
(and assuming no disturbances in equilibrium):

$$\begin{aligned}\mathbf{Births}_t =& \beta_0 \left(1 + \beta_2 + \beta_2^2 + \beta_2^3 + \cdots\right) + \\ & \beta_1 \left(\mathbf{Income}^\star + \beta_2 \mathbf{Income}^\star + \beta_2^2 \mathbf{Income}^\star + \cdots\right) + \\ =& \beta_0 \left(\frac{1}{\beta_2}\right) + \\ & \beta_1 \left(\frac{1}{\beta_2}\right) \mathbf{Income}^\star\end{aligned}$$

So long as $-1 < \beta_2 < 1$.[1]

1. This simplification comes from $\sum_{k=0}^{\infty} p^k = \frac{1}{p}$ for $-1 < k < 1$.